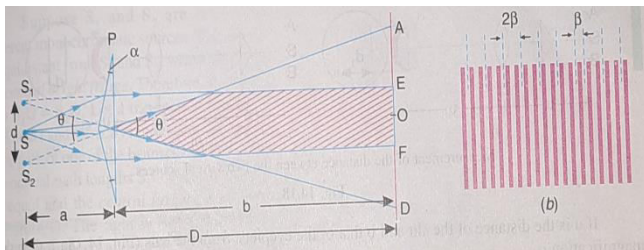
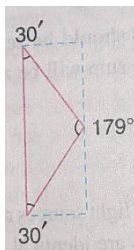


Fresnel's Biprism

It is an optical device which is used to produce the coherent sources of light by the phenomenon of refraction of light. Here we use division of wavefront method to produce two coherent sources of light.

The biprism consists of two prisms of very small refracting angles joined base to base. In practice, a thin glass plate is taken and one of its faces is ground and polished till a prism is formed with an obtuse angle of about 179° and two side angles of $30'$.



Experimental arrangement:

A monochromatic light source such as sodium vapour lamp illuminates a vertical slit S. The slit S acts as a narrow linear light source. The biprism is placed in such a way that its refracting edge is parallel to the length of the slit S. A single cylindrical wavefront impinges on both prisms. The top portion of the wavefront is refracted downward and appears to be emanated from the virtual source S_1 . The lower segment falling on the lower part of the prism, is refracted upward and appears to be emanated from the virtual source S_2 . The virtual sources S_1 and S_2 are coherent. The light waves from these two sources interfere in the region beyond the biprism and on the screen we get alternate bright and dark fringes.

Theory

If d is the distance between coherent sources S_1 and S_2 , D is the distance between coherent sources and screen and λ is the wavelength of light, then the fringe width is given by

$$\beta = \lambda D / d \quad \text{-----(i)}$$

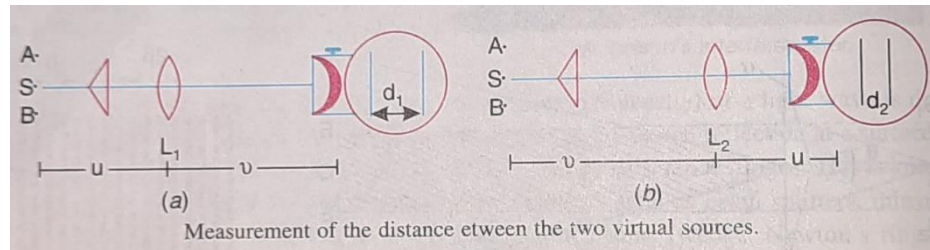
Determination of wavelength of light:

The slit S, the biprism and a micrometer eyepiece are mounted on the uprights and are adjusted to be at the same height in a straight line. The slit is made vertical and parallel to the refracting edge of the biprism. The slit is illuminated with the monochromatic light. The biprism is moved along the optical bench to get two equally bright vertical slit images. Then the eyepiece is moved till the fringes appear in the focal plane of the eyepiece.

(a) Determination of fringe width β :

When the fringes are observed in the field of view of eyepiece, the vertical cross wire is made to coincide with the center of one of the bright fringe. Its reading is taken. Let it be x_1 . Now the eyepiece is moved slowly to cross N bright fringes and the final reading is taken. Let it be x_2 . Then the fringe width is calculated as, $\beta = (x_2 - x_1)/N$.

(b) Determination of d :



A convex lens of short focal length is placed between the slit and the eyepiece without disturbing their positions. The lens is moved back and forth near the biprism till a sharp pair of images of the slit is obtained in the field of view of the eyepiece. The distance between the images is measured. Let it be d_1 .

$$\text{In this case, } v/u = d_1/d \text{ ----- (2)}$$

The lens is then moved to a position nearer to the eyepiece to get the pair of images again. The distance between the images is measured again. Let it be d_2 .

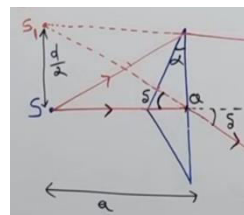
$$\text{In this case, } u/v = d_2/d \text{ ----- (3)}$$

$$\text{Multiplying (2) and (3), } 1 = d_1 d_2 / d^2$$

$$\text{Or, } d = \sqrt{d_1 d_2} \text{ ----- (4)}$$

Hence Knowing the values of $D = (a + b)$, d and β , the wavelength of monochromatic light can be found.

Aliter to find value of d :



$$\text{For a small angled prism, } \delta = \alpha (\mu - 1) \text{ ----- (5)}$$

where α is the refracting angle of the prism.

$$\text{From the figure, } \delta \approx \tan \delta = \Theta/2 = (d/2)/a = d/2a \text{ ----- (6)}$$

Now from (5) and (6),

$$\alpha (\mu - 1) = d/2a$$

or $d = 2a \alpha (\mu - 1).$